Reanalysis of Vascular Smooth Muscle Cell Phenotype Switching in Carotid Atherosclerosis

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Abstract. This report describes a reanalysis of the single-nucleus RNA-seq dataset associated with the study *Vascular smooth muscle cell phenotype switching in carotid atherosclerosis*. The goal was to reproduce the main cellular and molecular findings relevant to vascular smooth muscle cell (VSMC) phenotype switching, with emphasis on the HDAC9-associated mechanism proposed in the original paper. The analysis used the authors' raw count matrix and metadata, reconstructed a Seurat object, performed quality-control diagnostics, generated a batch-aware scVI latent representation, and examined cell-type composition, marker expression, VSMC-focused gene panels, and exploratory disease-control contrasts. The results reproduced the main structure of the dataset: major vascular, stromal, and immune cell populations were recovered, VSMCs formed the largest population, and disease samples showed expression patterns consistent with VSMC phenotypic modulation. HDAC9 and MALAT1 were broadly detectable, while immune-like markers such as CD68 and LGALS3 were strongest in macrophages but also detectable in disease-associated VSMCs. Gene Ontology gene set enrichment analysis of VSMC marker ranks showed enrichment for muscle cell differentiation, muscle contraction, muscle tissue development, and supramolecular fiber organization, supporting the interpretation that the VSMC compartment is defined by a differentiated contractile program that can be altered during phenotype switching. A specific quality-control issue was the presence of two RNA-complexity bands, estimated from residual gene detection after accounting for total UMI count. These bands were not removed globally because they were unevenly distributed across cell types; instead, they were retained as metadata and included as covariates in exploratory single-cell differential-expression tests. The main limitation is that condition and donor are confounded: the control condition is represented by one donor, whereas the disease condition is represented by two donors. Therefore, disease-control contrasts are interpreted as exploratory rather than definitive.

Keywords: single-nucleus RNA-seq · atherosclerosis · vascular smooth muscle cells · HDAC9 · Seurat · scVI

1 Introduction

Atherosclerotic carotid plaque disease is clinically important because plaque instability contributes to large-vessel ischemic stroke. The original paper by Chou and colleagues investigated whether vascular smooth muscle cells (VSMCs) participate in carotid plaque biology through HDAC9-associated phenotypic switching [1]. The central mechanism proposed by the authors is that HDAC9 modifies VSMC phenotype, promotes expression of immune-like markers, and contributes to recruitment of macrophages. This is important because inflammatory activity in plaques is often attributed mainly to immune cells, whereas VSMCs can also acquire macrophage-like features during atherogenesis.

The paper connects the single-nucleus RNA-seq data to a broader experimental model. In cell-culture experiments, lipid-treated VSMCs showed increased HDAC9, CD68, and LGALS3 expression and decreased contractile-gene expression, including ACTA2, SM22/TAGLN, and MYH11. Silencing HDAC9 reduced immune-marker expression and restored the contractile gene program. The authors also reported that conditioned medium from lipid-treated VSMCs promoted macrophage migration, and that SERPINE2/PN-1 was a candidate secreted mediator in this process [1]. Together, these results support a mechanism in which diseased VSMCs move away from a contractile phenotype and toward a phenotype with immune-like and secretory properties.

The purpose of this project was not to fully reproduce all experimental work in the paper, but to reanalyze the released single-nucleus RNA-seq dataset and evaluate whether the computational results are consistent with the proposed mechanism. The report focuses on four questions: (i) whether major cell types can be recovered from the data, (ii) whether VSMCs show expression patterns consistent with phenotype switching, (iii) how quality-control patterns, especially RNA-complexity bands, affect the interpretation, and (iv) how donor and biosample structure limit disease-control inference.

2 Materials and methods

2.1 Dataset and metadata

The analysis used the raw count matrix, gene file, barcode file, and metadata provided with the original study. The gene annotation file contained both Ensembl identifiers and gene symbols; gene symbols were used as Seurat feature names, while the raw count matrix was retained as the basis for normalization and downstream expression analysis. The metadata were aligned to the count matrix by cell barcode before object construction. This step is essential because direct assignment by row order can silently mismatch cells and metadata.

The released matrix contained 6,049 nuclei and 33,694 genes. The metadata included donor identifier, biosample identifier, assigned cell type, disease label, UMI count, detected gene count, mitochondrial fraction, exon proportion, entropy, and doublet score. The project defined a simplified condition variable, `condition2`, with two levels: `control` and `disease`. In this dataset, donor C7 represents the control condition, whereas donors 751 and 826 represent the disease condition. This structure is biologically useful for exploratory visualization, but it creates a major confounding problem for statistical interpretation because donor identity and disease status cannot be separated cleanly.

2.2 Initial object construction and normalization

The count matrix was loaded using `ReadMtx` with the second column of the gene file as the feature name. A Seurat object was then created using a minimal feature threshold. The object was normalized with log-normalization using a scale factor of 10,000. Highly variable genes were selected with the `vst` method. Genes often associated with technical effects, including mitochondrial, ribosomal, hemoglobin, and MALAT1 if present among variable genes, were removed from the highly variable gene set before dimensionality reduction.

2.3 Quality-control metrics and RNA-complexity bands

The original authors had already performed detailed sample-level and nucleus-level QC before releasing the dataset. According to the supplementary methods, they removed entire experiments that were strong QC outliers, then filtered droplets using per-experiment IQR-based cutoffs for UMI count, gene count, exon proportion, entropy, and doublet score. They also applied hard cutoffs requiring more than 500 UMIs, more than 150 genes, and less than 1% mitochondrial reads. After these filters, 6,049 nuclei remained [2].

In the present reanalysis, the QC step was therefore diagnostic rather than primarily exclusionary. The code recalculated or imported the following quantities: `nCount_RNA`, `nFeature_RNA`, mitochondrial percentage, and an RNA-complexity metric. A simple ratio,

$$\text{complexity}_i = \frac{\text{nFeature\_RNA}_i}{\text{nCount\_RNA}_i},$$

was calculated, but the main complexity measure used in the workflow was a residual score. A linear model was fitted:

$$\log_{10}(\text{nFeature\_RNA}_i) = \alpha + \beta \log_{10}(\text{nCount\_RNA}_i) + \varepsilon_i.$$

The residual ε_i was stored as `complexity_resid`. This residual represents whether a nucleus has more or fewer detected genes than expected for its total UMI count. Positive residuals correspond to higher-than-expected gene complexity, whereas negative residuals correspond to lower-than-expected gene complexity.

The residual distribution was separated into two groups using `kmeans` with two centers. The cluster with the lower mean residual was labeled `lower_complexity`; the other was labeled `higher_complexity`. This approach is preferable to a simple threshold on `nFeature_RNA` or `nCount_RNA`, because it distinguishes nuclei with low gene diversity relative to their sequencing depth from nuclei that simply have low read depth overall.

The complexity bands were not used as a global cell filter. This decision was made because the low-complexity band was unevenly distributed across cell types. For example, the lower-complexity fraction was approximately 31.9% in VSMCs, 52.7% in lymphocytes, 27.9% in endothelial cells, 23.4% in macrophages, 13.5% in pericytes, and only 5.6% in fibroblasts. Removing all low-complexity nuclei would therefore have distorted the biological composition of the dataset. Instead, `qc_band` and `complexity_resid` were retained as metadata. The residual score was also included as a latent variable in exploratory disease-control differential-expression tests.

2.4 Batch-aware embedding, donor control, and clustering

Although the downloaded matrix already contained both control and disease nuclei, it was not assumed to be pre-integrated. A combined count matrix is not the same as an integrated latent representation. The original supplementary methods state that the authors used the top 2,000 highly variable genes followed by scVI batch correction for each experiment, with 50 latent dimensions, a neighborhood graph based on scVI embeddings, UMAP visualization, and Leiden clustering [2,4].

The project followed the same general strategy. A precomputed `latent.RData` file was generated using scVI through `reticulate`, with `biosample_id` as the batch key. This variable was chosen because the paper describes correction per experiment, and `biosample_id` is the closest available experiment/library-level variable in the metadata. The scVI latent representation was used for nearest-neighbor graph construction, Leiden clustering, and UMAP visualization. It was not used as the expression matrix for differential expression. Expression plots and differential-expression tests were based on the RNA assay.

This distinction is important. Batch correction is useful for visualization and clustering, but it does not solve the biological limitation that one donor represents the control condition and two donors represent the disease condition.

2.5 Marker panels and exploratory differential expression

Cell-type identity was checked with canonical marker panels for endothelial cells, VSMCs, fibroblasts, pericytes, macrophages, and lymphocytes. The main biological analysis focused on the mechanism described in the original paper. Gene panels were organized into: an HDAC9 axis (HDAC9, MALAT1, SMARCA4/BRG1, SERPINE2/PN-1), a VSMC contractile program (ACTA2, TAGLN/SM22, MYH11, CNN1, SMTN), immune-like markers (CD68, LGALS3, LYZ, MSR1, APOE), and endothelial control markers.

Exploratory disease-control tests were run within selected cell types using Seurat's `FindMarkers` method. The model included `nCount_RNA`, `percent.mt`, and `complexity_resid` as latent variables. These tests were used to nominate candidate genes and guide visualization.

2.6 Gene set enrichment analysis

To connect the gene-level results with biological processes, Gene Ontology (GO) gene set enrichment analysis was added to the workflow. For each annotated cell type, the marker table from `FindAllMarkers` was used to create a ranked gene list, with genes ordered by `avg_log2FC`. Enrichment was then performed with `clusterProfiler::gseGO`, using human gene symbols, the `org.Hs.eg.db` annotation package, all GO domains, 10,000 permutations, and gene-set size limits of 3 to 800 genes [5,6,7]. The VSMC analysis was the focus of interpretation because the original paper centers on VSMC phenotype switching.

The enrichment results were interpreted qualitatively. In the project code, `pAdjustMethod = "none"` was used, so the enrichment plots are best treated as exploratory process-level summaries. In addition, the VSMC marker table mostly contains genes positively enriched in VSMCs compared with other cell types. Therefore, the `activated` and `suppressed` labels in the enrichment plot should be read as enrichment toward the high-ranked or low-ranked end of the VSMC marker list. Disease-control interpretation was instead supported by donor-level dotplots and exploratory within-VSMC differential-expression results.

3 Development of modeling

The computational workflow was designed as a model of the paper's proposed mechanism. The biological hypothesis can be summarized as follows: in carotid atherosclerosis, VSMCs shift away from a contractile phenotype, show increased HDAC9-associated activity, express immune-like markers such as CD68 and LGALS3, and may secrete factors such as SERPINE2/PN-1 that contribute to macrophage recruitment.

The analysis tested this mechanism in three layers. First, cell-type structure was examined to verify that the dataset contained the expected vascular and immune populations. Second, expression of paper-focused genes was compared across cell types and conditions. Third, VSMC-specific plots were used to ask whether candidate disease-associated expression patterns were consistent across disease donors or mostly donor-specific.

The most important design choice was to separate the integration task from the differential-expression task. The scVI latent space was used to make the dataset interpretable as a single map, while the RNA assay was used to inspect gene expression. This avoids a common problem in single-cell analysis: using an integrated embedding to visualize cells is appropriate, but using corrected embeddings as expression values for differential expression can remove or distort biological differences.

4 Results

4.1 The integrated UMAP recovered major cell populations

scVI-based UMAP separated the expected major cell types, including VSMCs, endothelial cells, macrophages, lymphocytes, fibroblasts, and pericytes (Fig. 1).

The donor panel showed that donor identity did not completely dominate the global embedding, although donor-specific enrichment was still visible in some regions. The condition panel showed the expected strong imbalance between control and disease nuclei because the control condition was represented only by donor C7, while the disease condition was represented by donors 751 and 826. The QC-band panel showed that lower-complexity nuclei were present in the map but did not define all major cell-type clusters.

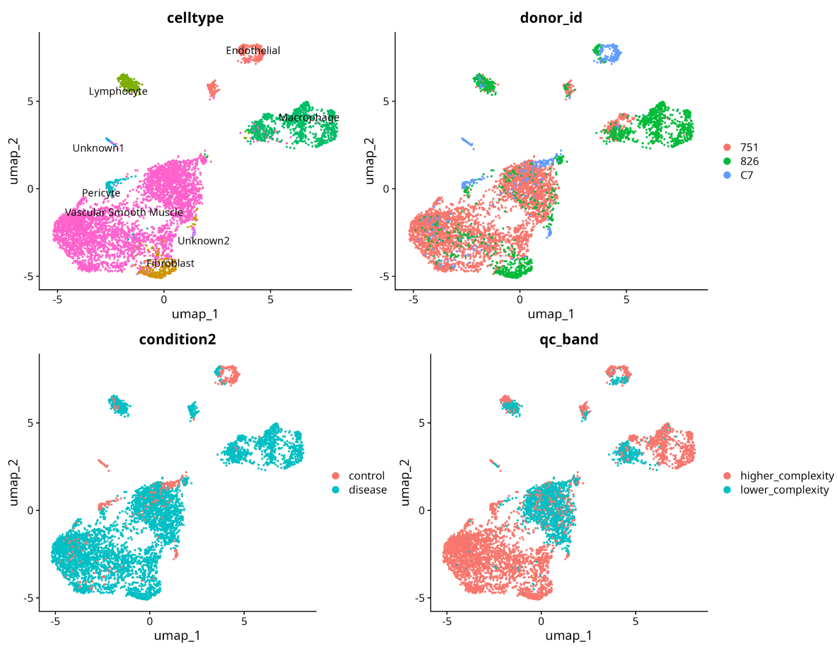

Fig. 1. UMAP overview of the scVI latent space. The panels show cell-type labels, donor identity, disease condition, and RNA-complexity band. Cell-type structure is clear, while donor and condition remain important interpretive variables.

4.2 Cell-type composition differed strongly by donor

Cell-type composition varied substantially across donors (Fig. 2). Donor 751 was dominated by VSMCs, which represented approximately 90.6% of nuclei from that donor. Donor 826 was more macrophage-rich, with macrophages representing approximately 47.0% of nuclei and VSMCs only 18.8%. The control donor C7 contained a more mixed population, with endothelial cells and VSMCs each representing roughly one third of nuclei.

This result is important for interpretation. Disease-control expression differences can reflect true disease biology, but they can also reflect differences in donor composition, biosample processing, or the specific tissue region captured.

Therefore, cell-type-specific and donor-level visualizations are essential before interpreting disease-associated genes.

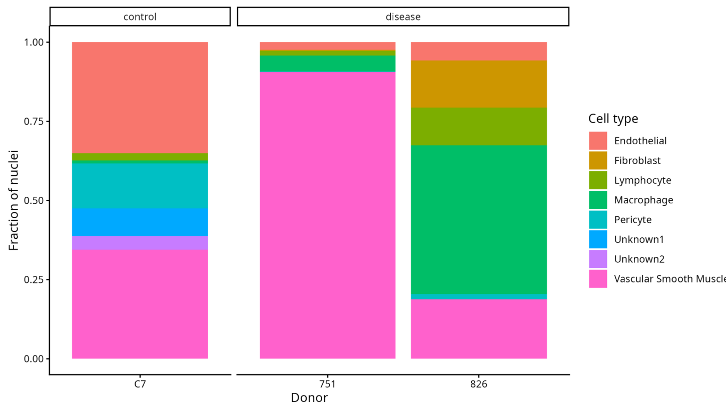

Fig. 2. Cell-type composition by donor. The two disease donors have very different cellular composition: donor 751 is VSMC-rich, whereas donor 826 is macrophage-rich. This reinforces the donor/condition confounding limitation.

4.3 QC diagnostics revealed two RNA-complexity modes

The QC violin plots showed that UMI count, detected gene count, mitochondrial percentage, and complexity residuals differed across nuclei but did not by themselves justify broad filtering by condition (Fig. 3). Mitochondrial content was very low overall, consistent with the fact that the authors had already applied strict mitochondrial filters before data release.

The scatter plot of `nCount_RNA` versus `nFeature_RNA` revealed two clear complexity bands (Fig. 4). Both bands showed the expected positive relationship between UMI count and detected genes, but the lower band had fewer detected genes for the same UMI count. This indicates lower gene complexity, not simply lower sequencing depth. The histogram of residual complexity confirmed the bimodal structure used to define `higher_complexity` and `lower_complexity` classes (Fig. 5).

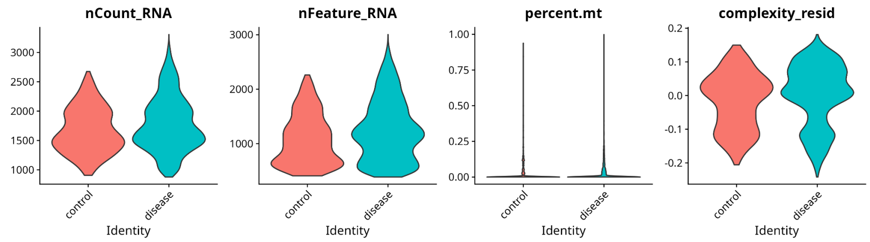

Fig. 3. QC metrics by condition. The plots summarize UMI counts, detected genes, mitochondrial percentage, and residual complexity. The mitochondrial signal is low, and complexity variation is visible in both conditions.

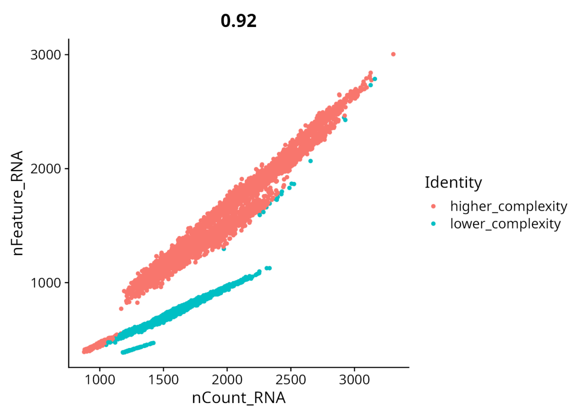

Fig. 4. Relationship between total UMI count and detected genes, colored by RNA-complexity band. The lower-complexity band represents nuclei with fewer detected genes than expected for their UMI count.

Because the lower-complexity group was not evenly distributed across cell types, it was not removed globally. This is a central QC decision in the project. A global filter would have removed many lymphocytes and VSMCs while barely affecting fibroblasts, thereby changing the biological composition of the dataset. Instead, the complexity band was treated as a technical/biological covariate to be monitored. The residual score was incorporated into exploratory differential-expression models, and the band was shown in UMAP plots as a quality-control layer.

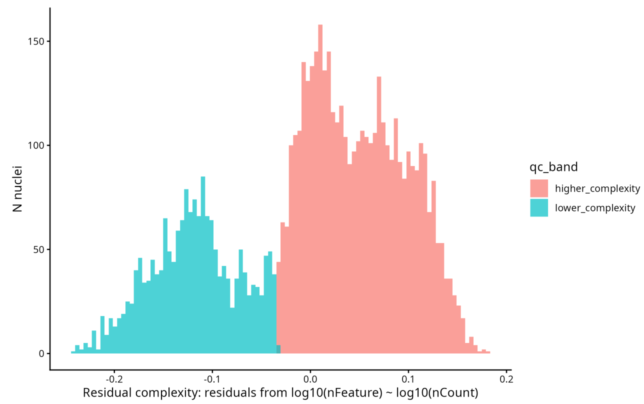

Fig. 5. Distribution of residual RNA complexity. Residuals were computed from the regression of $\log_{10}$(nFeature_RNA) on $\log_{10}$(nCount_RNA), and `kmeans` clustering with two centers was used to assign complexity bands.

4.4 Cell-type annotation was consistent with canonical markers

The annotation marker dotplot supported the major cell-type labels (Fig. 6). Endothelial cells expressed markers such as PECAM1, VWF, CDH5, and KDR/FLT1/TEK. VSMCs expressed ACTA2, TAGLN, MYH11, CNN1, and SMTN. Fibroblasts showed extracellular matrix markers such as DCN, LUM, COL1A1, COL1A2, COL6A3, and MFAP5. Macrophages showed LYZ, CD68, CD163, MSR1, MRC1, and complement genes, while lymphocytes showed PTPRC, CD2, CD3D, CD3E, TRAC, IL7R, and BCL11B.

The plot also illustrates a point emphasized by the original paper: vascular wall cell types are not always separated by perfectly exclusive markers. VSMCs, pericytes, and fibroblasts share matrix and mural-cell programs, and immune-like markers can be detected outside the macrophage cluster.

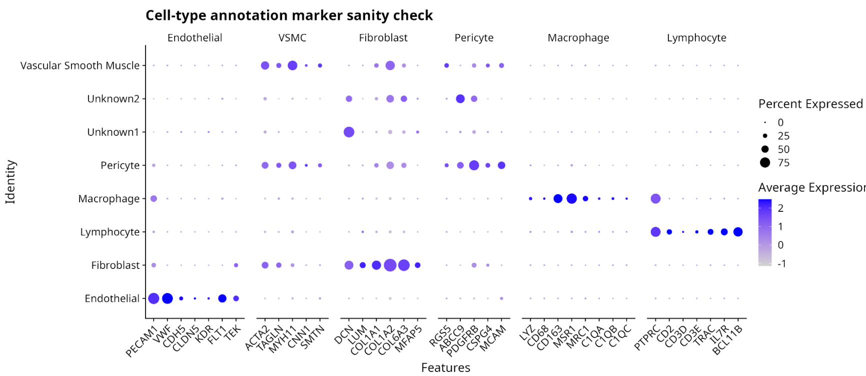

Fig. 6. Cell-type annotation marker sanity check. The expected marker programs are enriched in the corresponding annotated cell types, supporting the metadata-based cell-type labels used in the reanalysis.

4.5 Paper-focused gene panels support the VSMC phenotype-switching narrative

The HDAC9/VSMC phenotype panel summarized the main mechanism described in the original paper (Fig. 7). MALAT1 was broadly detected, which is expected for a highly expressed nuclear long non-coding RNA. HDAC9 was detected across several cell types, with particular interest in VSMCs. VSMC contractile genes, including ACTA2, TAGLN, MYH11, CNN1, and SMTN, were enriched in VSMCs and pericyte-like mural populations. Immune-like markers were strongest in macrophages, but CD68 and LGALS3 showed detectable signal in VSMCs and other vascular cells, especially in disease-associated groups.

This result is consistent with the paper's interpretation, but with an important nuance. The single-nucleus dataset does not by itself prove that VSMCs have fully transdifferentiated into macrophage-like cells. Instead, it supports a more cautious statement: disease-associated VSMCs show partial expression of genes related to the HDAC9 axis and immune-like activation, while retaining components of the contractile VSMC identity. This is compatible with a phenotype-switching model.

The feature plots for key genes provided spatial confirmation on the UMAP (Fig. 8). ACTA2 and MYH11 were concentrated in the VSMC-rich region, while CD68 and LGALS3 were strongest in macrophage regions with additional low-level signal elsewhere. HDAC9 was more broadly distributed, which matches the need to interpret it in the context of cell type and phenotype rather than as a simple binary marker.

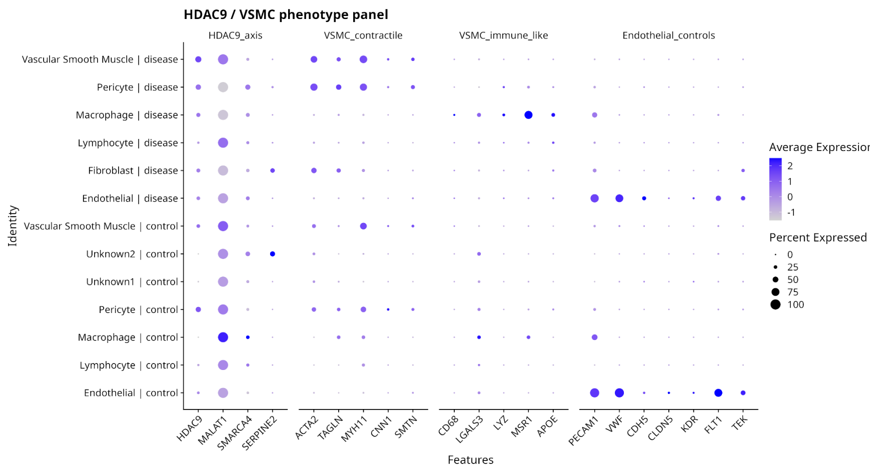

Fig. 7. Paper-focused HDAC9/VSMC phenotype panel. Gene groups summarize the HDAC9 axis, contractile VSMC identity, immune-like markers, and endothelial controls across cell type and condition.

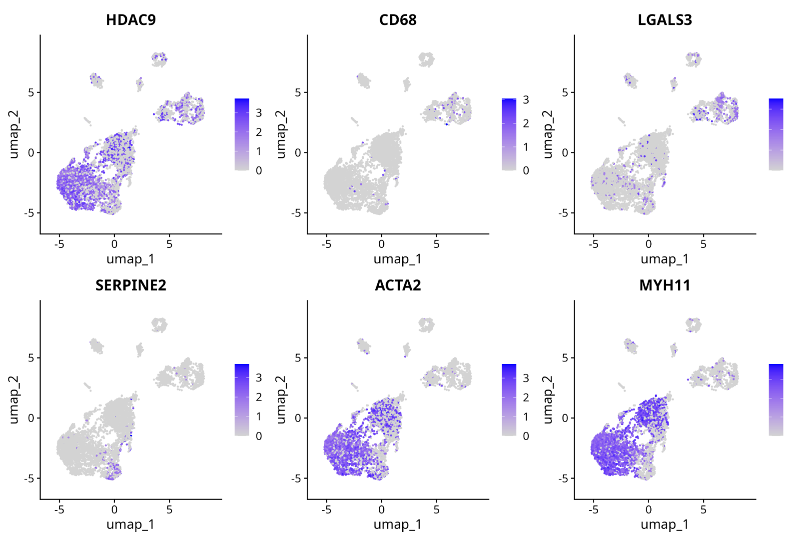

Fig. 8. UMAP feature plots for key genes in the proposed mechanism. Contractile genes localize mainly to the VSMC region, whereas immune-like markers are strongest in macrophages but are not completely restricted to them.

4.6 Gene set enrichment linked VSMC markers to muscle differentiation

The VSMC-focused gene set enrichment analysis connected the gene-level marker results to broader cell differentiation ontology. The most relevant activated terms were *muscle cell differentiation*, *muscle structure development*, *muscle tissue development*, *muscle contraction*, *muscle system process*, *supramolecular fiber*, and *cell periphery* (Fig. 9). These categories are consistent with the expected identity of differentiated VSMCs, whose biological role depends on cytoskeletal organization, contractility, and maintenance of mural-cell structure. This result supports the annotation of the VSMC compartment and provides a link to the original paper's mechanism: phenotype switching in disease can be understood as a shift away from a stable contractile/muscle differentiation program and toward immune-like or secretory behavior.

The gene-level results also fit this interpretation. Within the VSMC marker table, classical contractile and VSMC-associated genes such as *MYH11*, *ACTA2*, *MYLK*, *TAGLN*, *MYOCD*, *SPEG*, *PDE4D*, *ANK3*, *LAMA2*, and *HDAC9* were enriched in the VSMC compartment. Several of these genes correspond directly to the mechanism discussed in the original paper: *ACTA2*, *TAGLN*/SM22, and *MYH11* represent the contractile VSMC program, while *HDAC9* is the regulatory axis of interest. The disease-control candidate genes within VSMCs were more heterogeneous. Disease-higher candidates included genes such as *FHL5*, *LRRC17*, *FOSB*, *CYR61*, *SLC14A1*, *DDAH1*, *ITGBL1*, and *COL3A1*, suggesting changes in signaling, extracellular matrix organization, and cell-state remodeling. Control-higher candidates included genes such as *SYTL3*, *LRRC3B*, *CYP1B1-AS1*, *PTPRR*, *BHLHE40*, and *CLSTN2*. Because control is represented by a single donor, these genes should be treated as exploratory rather than confirmed repressed disease genes.

The ridge plot gives a complementary view of the same VSMC enrichment result (Fig. 10). The distributions for muscle-related terms are shifted toward higher enrichment values, indicating that these ontologies are supported by multiple genes rather than by a single marker. This strengthens the conclusion that the VSMC population is not identified only by one or two canonical genes, but by a coordinated contractile and differentiation-associated program.

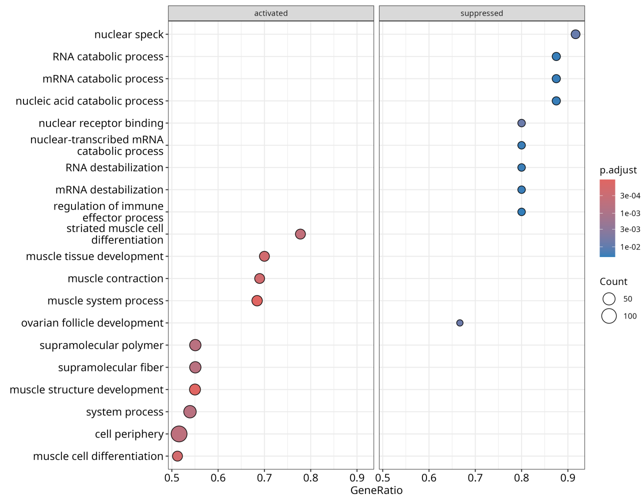

Fig. 9. Exploratory GO gene set enrichment analysis for VSMC marker ranks. High-ranked VSMC markers were enriched for muscle differentiation, muscle contraction, muscle tissue development, and structural fiber organization.

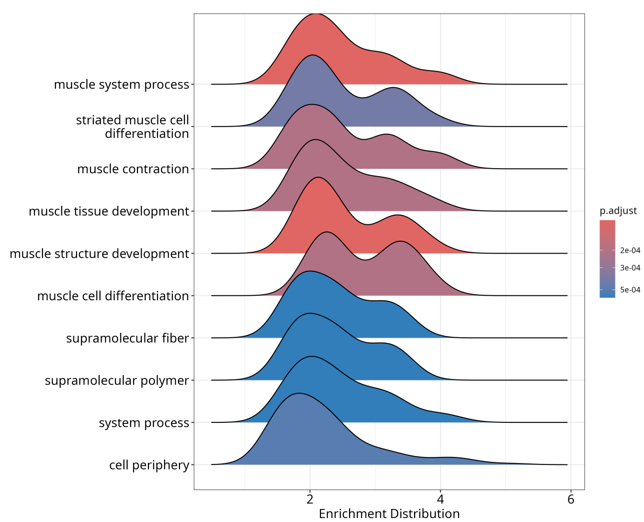

Fig. 10. Ridge plot of VSMC-enriched GO terms. Muscle system process, muscle contraction, muscle tissue development, muscle structure development, and muscle cell differentiation show enrichment distributions consistent with a coordinated VSMC differentiation program.

4.7 VSMC subclustering was informative but limited

The original paper reported VSMC subpopulations distinguished partly by PDE4D, ANK3, and LAMA2, but also stated that interpretation was limited by sample number. The reanalysis reached a similar conclusion. VSMC subclustering identified structure within the VSMC compartment, but the result was influenced by donor, condition, and QC band (Fig. 11). Therefore, VSMC subclusters are useful for exploratory visualization, but they should not be overinterpreted as stable biological states in this small dataset.

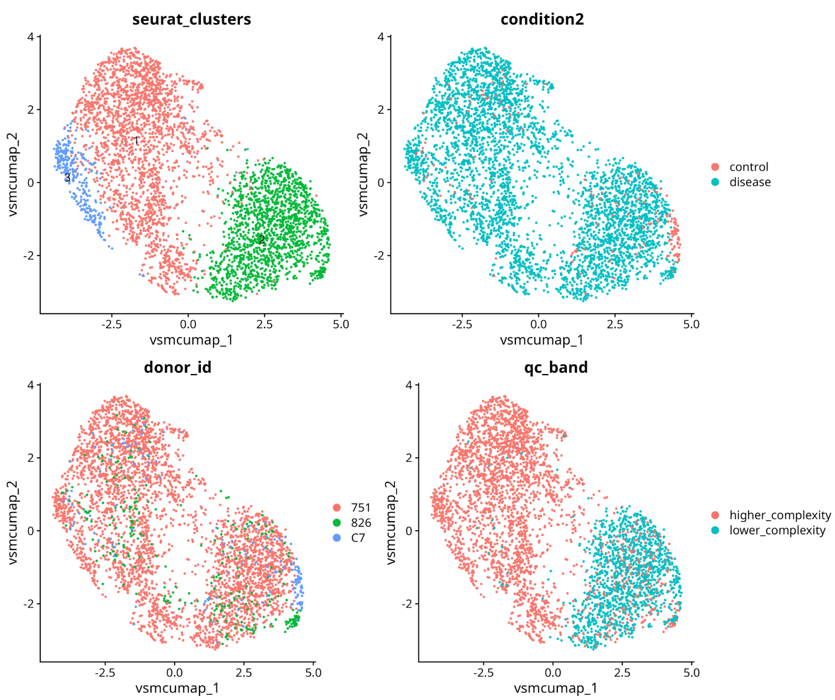

Fig. 11. VSMC-focused subclustering overview. VSMC substructure is visible, but condition, donor, and complexity band all contribute to the pattern. This supports cautious interpretation of VSMC subclusters.

4.8 Exploratory disease-control genes were checked at donor level

The VSMC donor-level dotplot was used to inspect whether exploratory candidate genes were consistent across disease donors (Fig. 12). This is essential because donor and condition are confounded. A convincing disease-associated candidate would be low in the control donor and high in both disease donors; a donor-specific candidate would be high in only one disease donor.

The plot showed that some exploratory disease-higher and control-higher genes followed the expected condition direction, but many patterns were donor-dependent. For example, several disease-higher candidates were stronger in donor 751 than in donor 826. Similarly, control-higher candidates were mostly strongest in C7, but this cannot be separated from the fact that C7 is the only control donor. These results are useful for hypothesis generation but should not be treated as validated disease biomarkers.

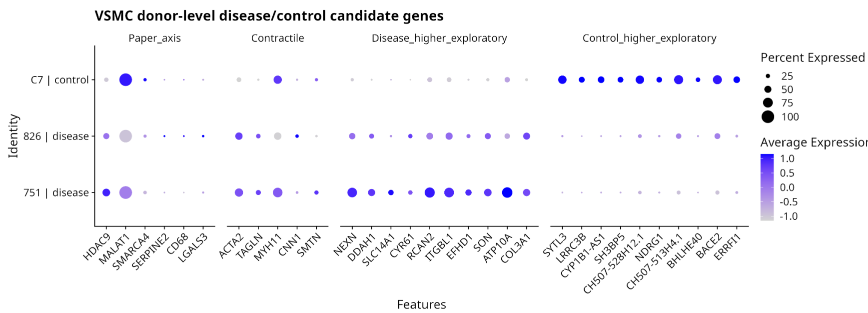

Fig. 12. VSMC donor-level disease/control candidate genes. Candidate expression was inspected separately by donor to evaluate whether disease-associated patterns were consistent across both disease donors or mainly donor-specific.

5 Conclusions

This reanalysis reproduced the main qualitative structure of the original carotid single-nucleus RNA-seq dataset. Major vascular and immune cell populations were recovered, and VSMCs formed a central population in the integrated map. The expression patterns of HDAC9, MALAT1, SERPINE2, contractile VSMC genes, and immune-like markers were broadly consistent with the mechanism proposed in the original paper: VSMCs in carotid disease can show features of phenotype switching, including partial immune-like activation and altered contractile identity. The added GO enrichment analysis supported this interpretation at the process level by showing that VSMC marker ranks were enriched for muscle cell differentiation, muscle contraction, muscle tissue development, and supramolecular fiber organization.

The QC analysis identified two RNA-complexity bands. These were estimated using residuals from the relationship between detected genes and total UMI count, then classified with two-group clustering. The lower-complexity band was not removed globally because it was unevenly distributed across biological cell types and would have biased cell-type composition. Instead, it was retained as a metadata field and incorporated as a covariate in exploratory differential-expression tests.

The donor and biosample structure is the main limitation of the analysis. Although the scVI latent embedding controlled for biosample-level effects during visualization and clustering, the dataset contains one control donor and two disease donors. Therefore, disease and donor effects remain confounded. The appropriate interpretation is that the computational results support the biological direction of the original paper, especially the HDAC9-associated VSMC phenotype-switching model, but disease-control gene-level differences should be considered exploratory and hypothesis-generating.

Data Availability

The single-nucleus RNA sequencing data analyzed in this project were obtained from the Single Cell Portal SCP2019 study page. Only the raw expression data were used in this reanalysis; the normalized expression matrix provided by the original authors was not used.

For convenience and reproducibility, a copy of the raw expression files used in this report is also provided in the project repository under `expression/raw`.

Code Availability

All code used for this reanalysis was written for this project and is publicly available on GitHub:

`https://github.com/dvgodoy/VSMC_Reanalysis`

The repository contains the R analysis workflow used to load the raw count matrix and metadata, construct the Seurat object, perform quality-control diagnostics, incorporate the scVI latent embedding for batch-aware clustering and UMAP visualization, generate the marker and differential-expression summaries, and produce the figures included in this report.

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